Computer Vision

DPS920

Session 21

Overview

Fire detection with neural networks
NN for classification vs Regression
Encoding overview
Softmax
Cross entropy loss

Agenda

Convolutional Neural Networks
Conv layer
Padding
Stride
Pooling
Code

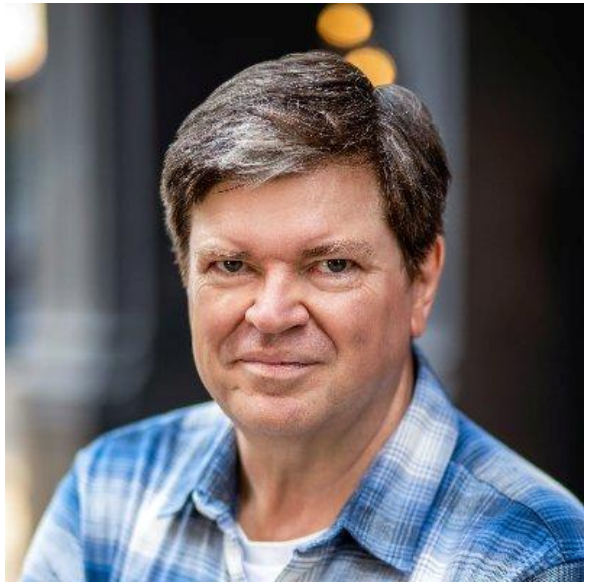
What is Left?

1. Optimization and Loss Function
2. Code + Logistic Regression
3. ML and Images
4. Perceptron and Neural Networks
5. Neural Networks
6. Convolution Neural Networks (CNN)
7. Advanced CNNs
8. Introduction to object detection, Segmentation and image generation methods with AI

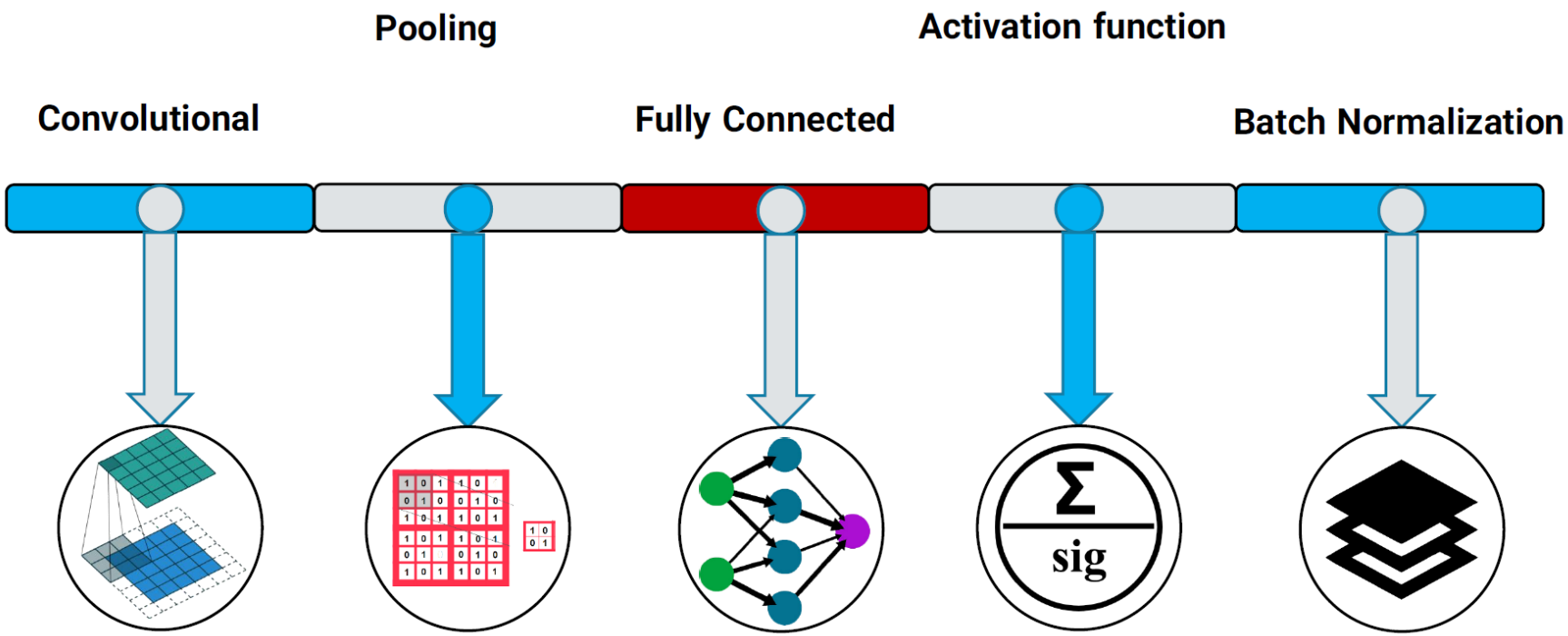
Convolutional Neural Networks (CNNs)

- Images are 2D or 3D grids with local dependencies
- Traditional NNs ignore locality and translation invariance
- We need a method that captures patterns like edges, textures, shapes

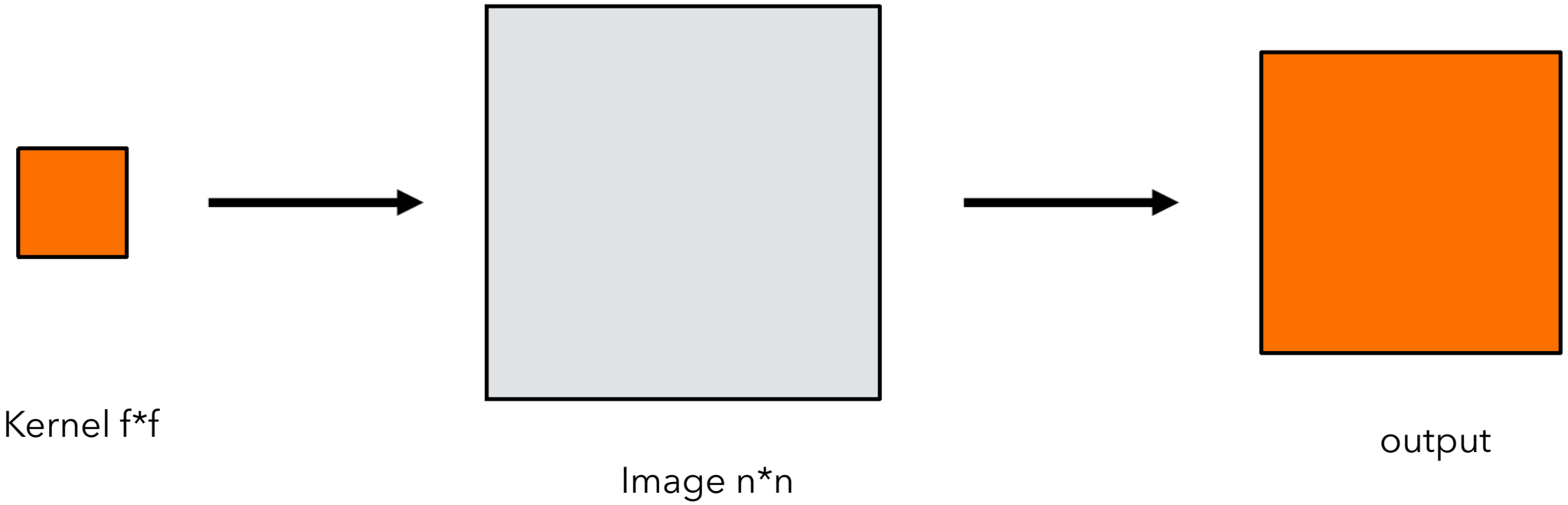
- LeNet: https://www.youtube.com/watch?v=FwFduRA_L6Q
- Convolution with Optimization



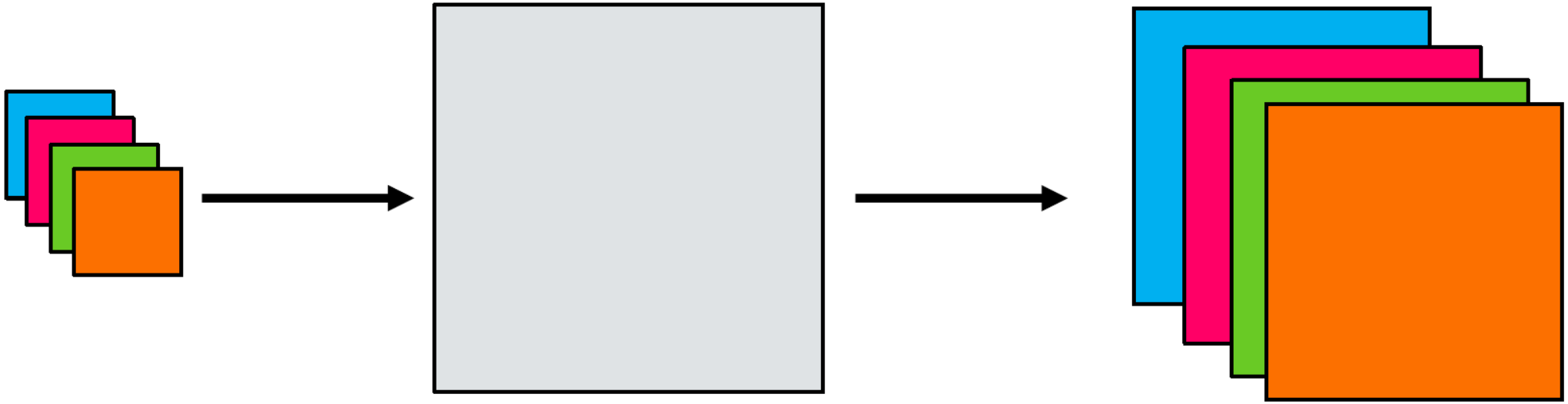
Layers in CNNs



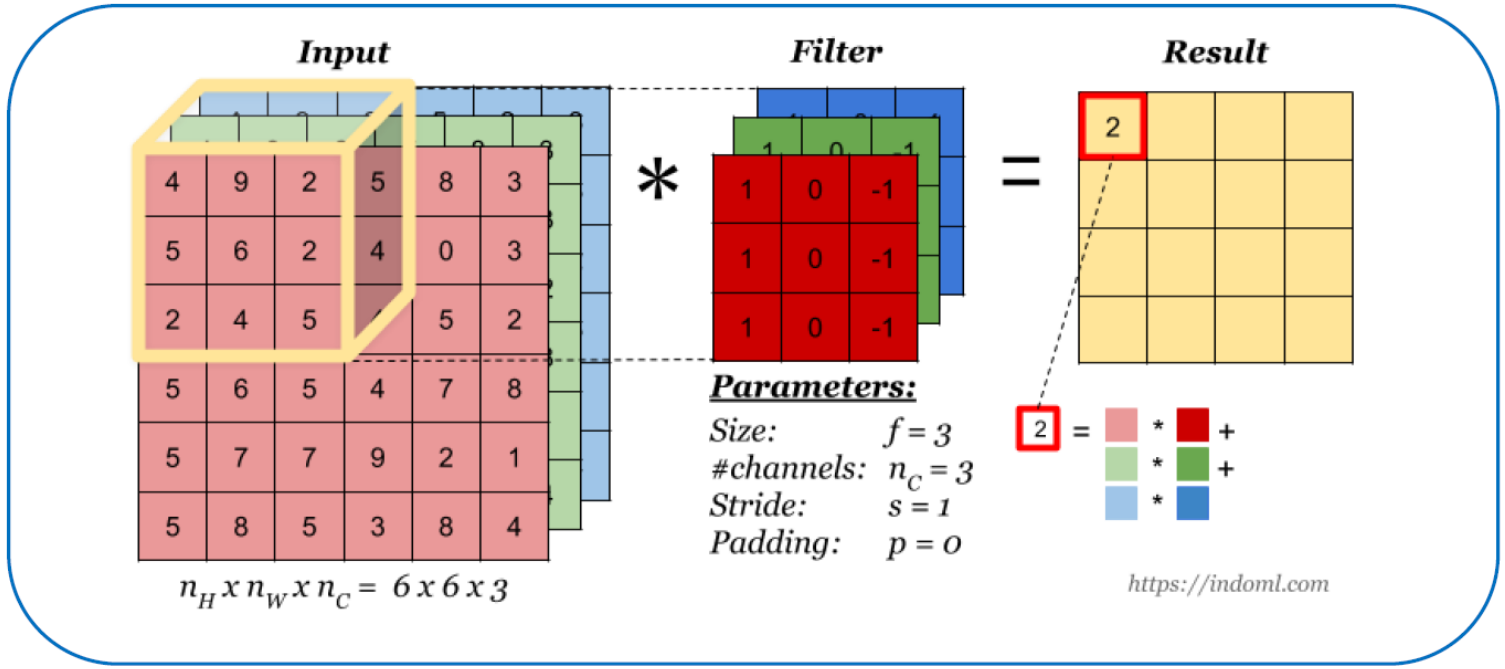
Convolution Layer



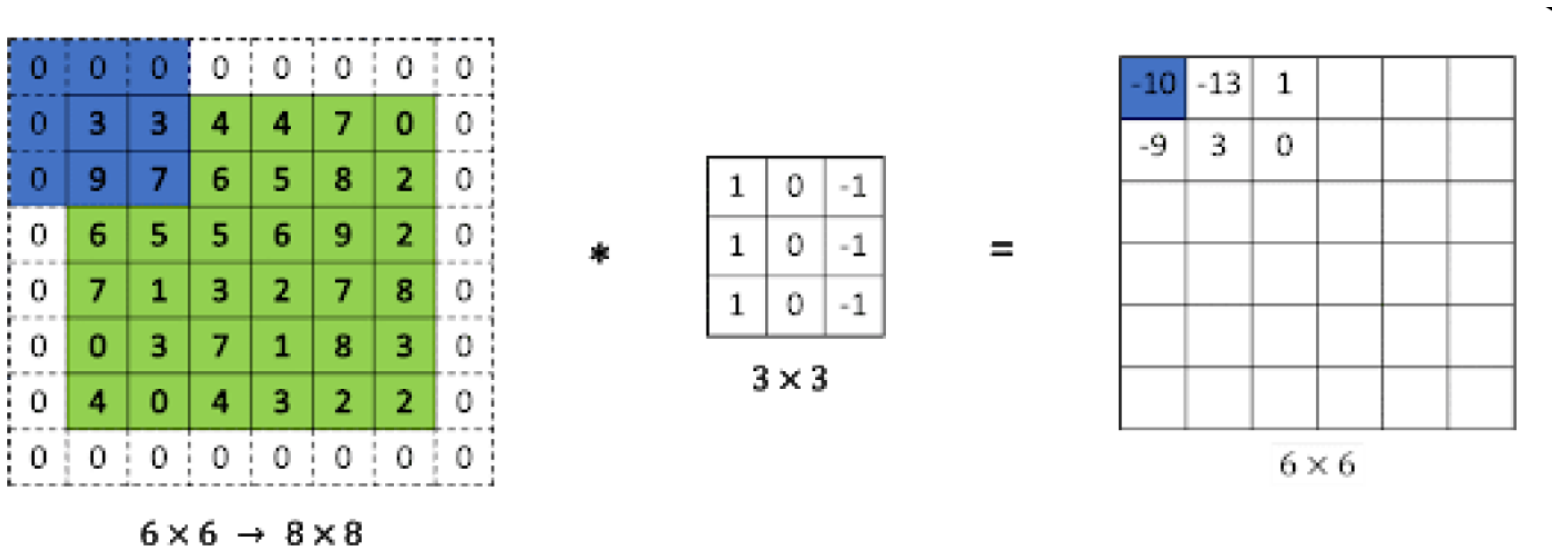
Multiple Convolutions



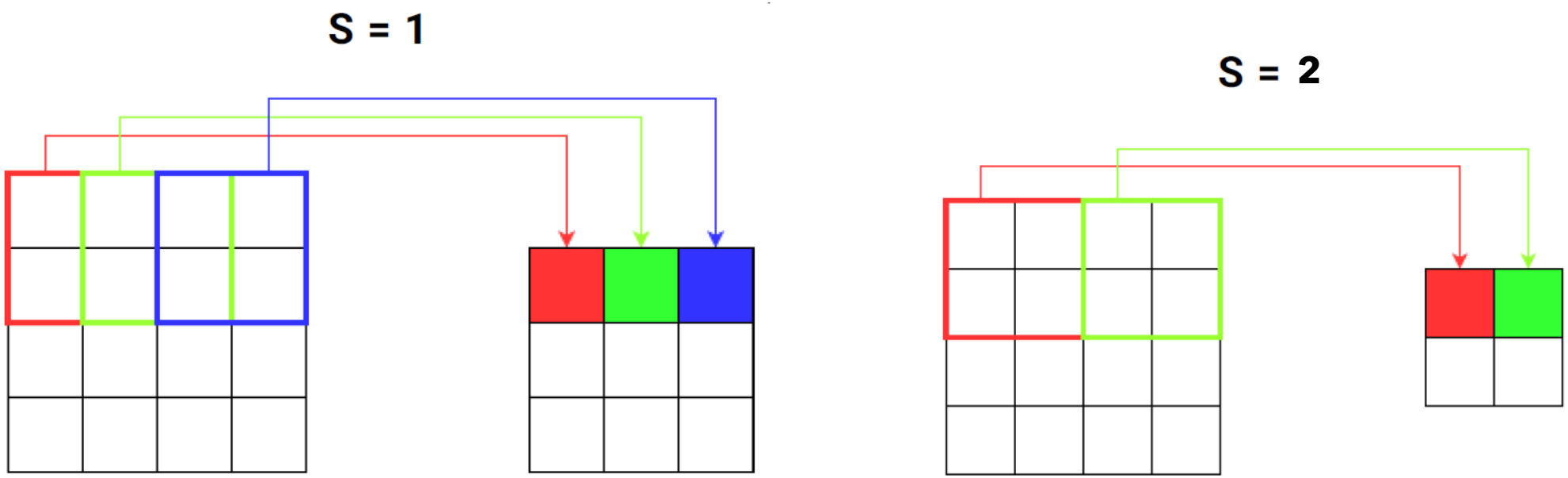
Convolution for Colored Images



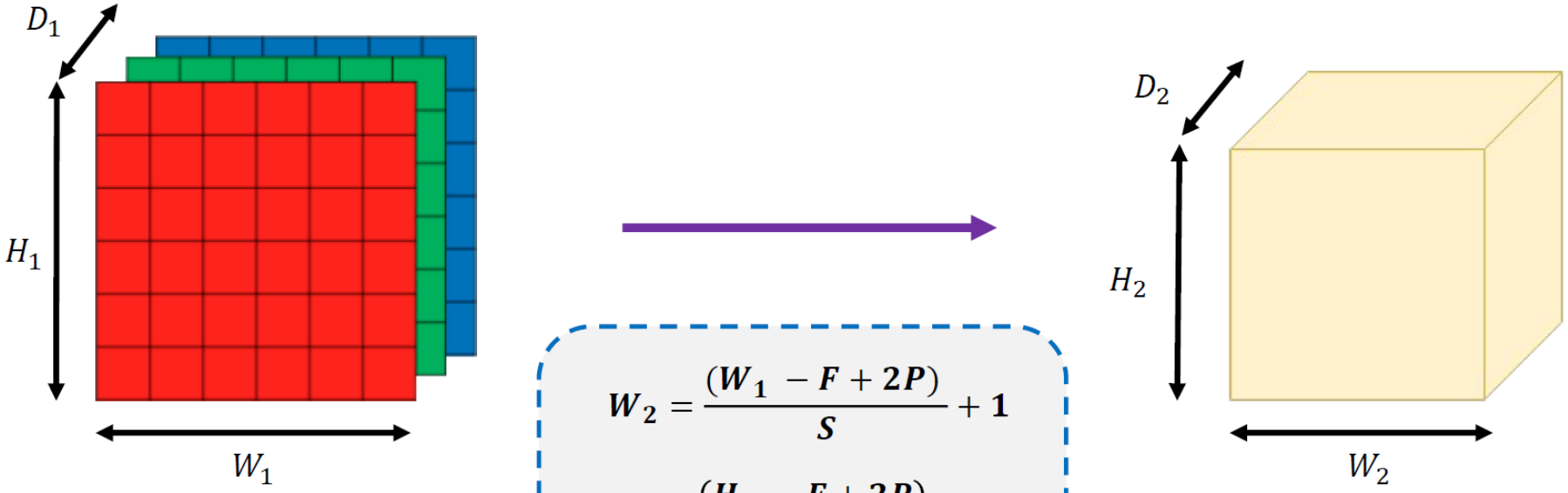
Padding


0 0 0 0 0 0 0 0
0 3 3 4 4 7 0 0
0 9 7 6 5 8 2 0
0 6 5 5 6 9 2 0
0 7 1 3 2 7 8 0
0 0 3 7 1 8 3 0
0 4 0 4 3 2 2 0
0 0 0 0 0 0 0 0
6 × 6 → 8 × 8
*
1 0 -1
1 0 -1
1 0 -1
3 × 3
=
-10 -13 1
-9 3 0
6 × 6

Stride



Post Convolution Dimensions

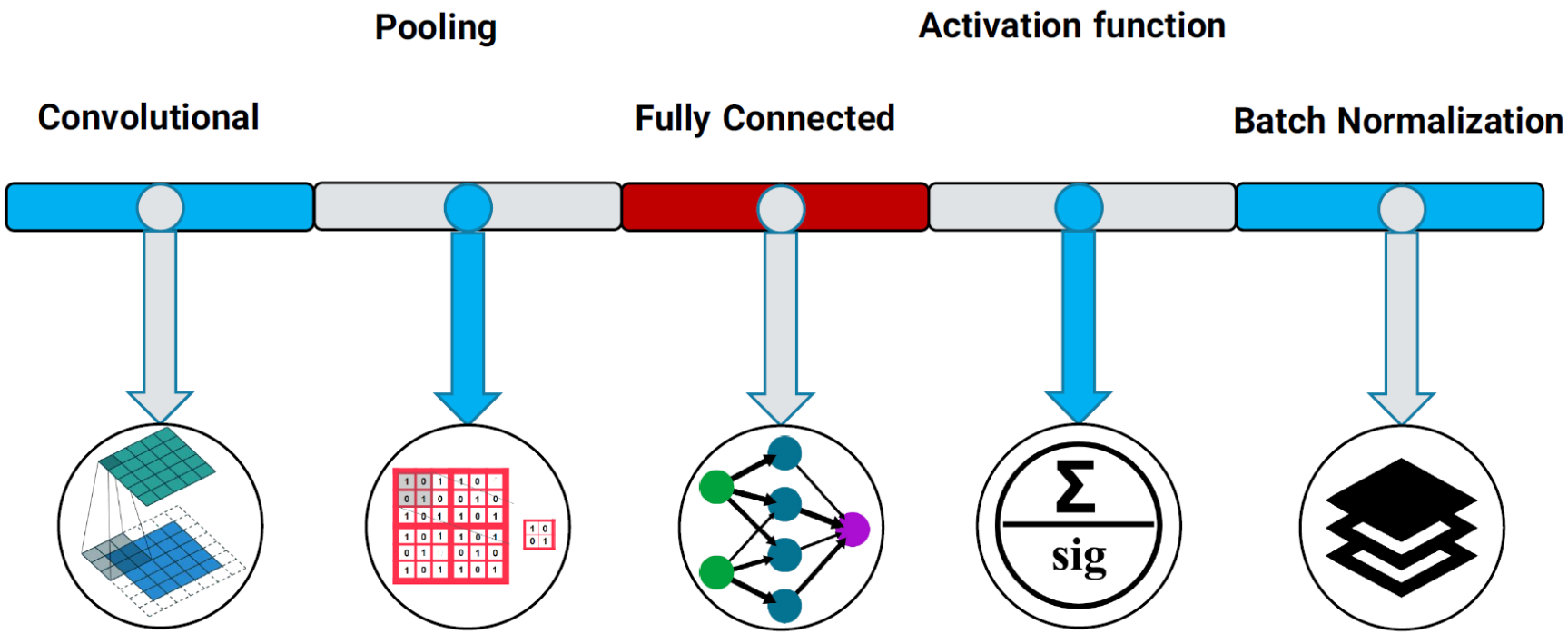


$$W_2 = \frac{(W_1 - F + 2P)}{S} + 1$$

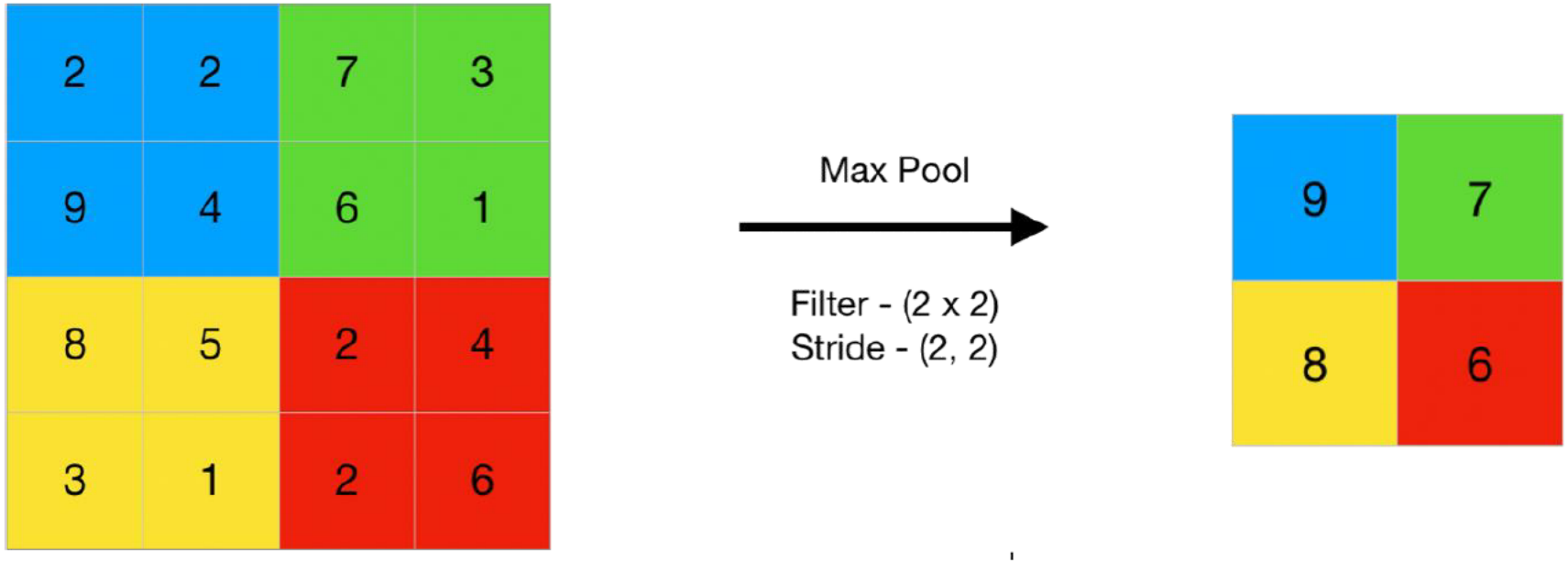
$$H_2 = \frac{(H_1 - F + 2P)}{S} + 1$$

$$D_2 = K$$

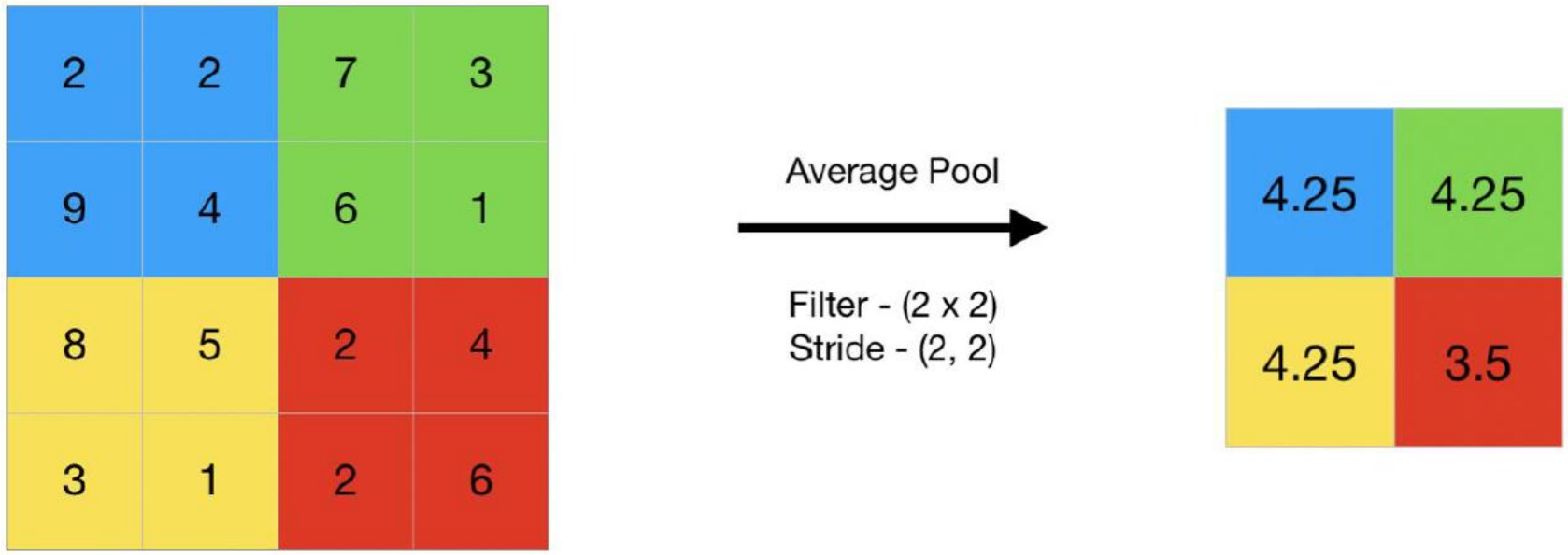
Layers in CNNs



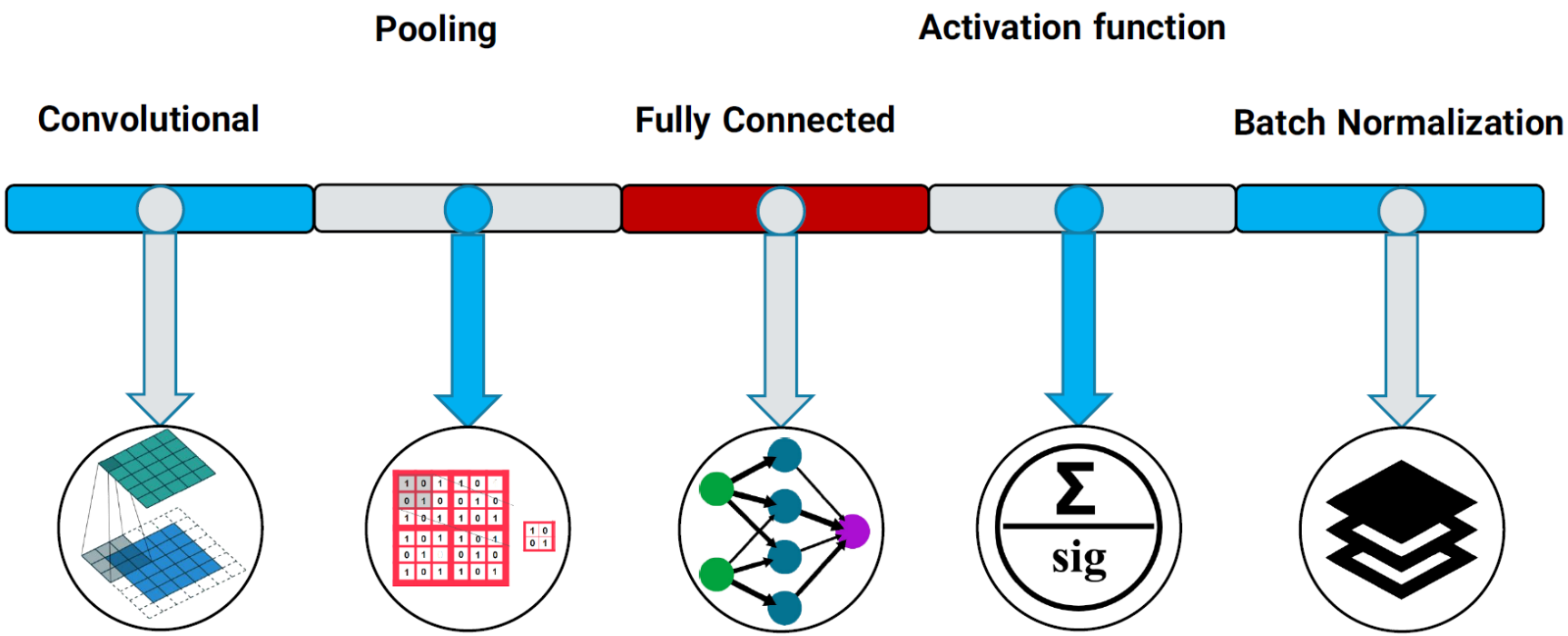
MaxPooling



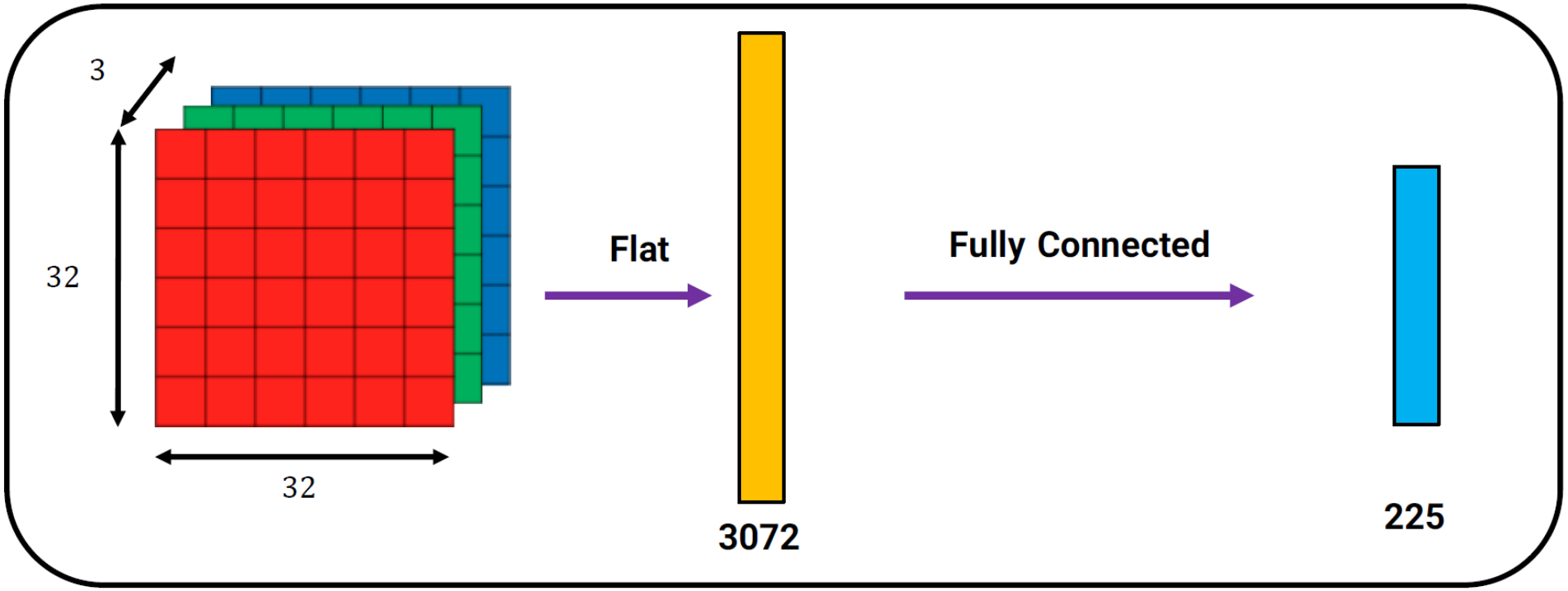
Average Pooling



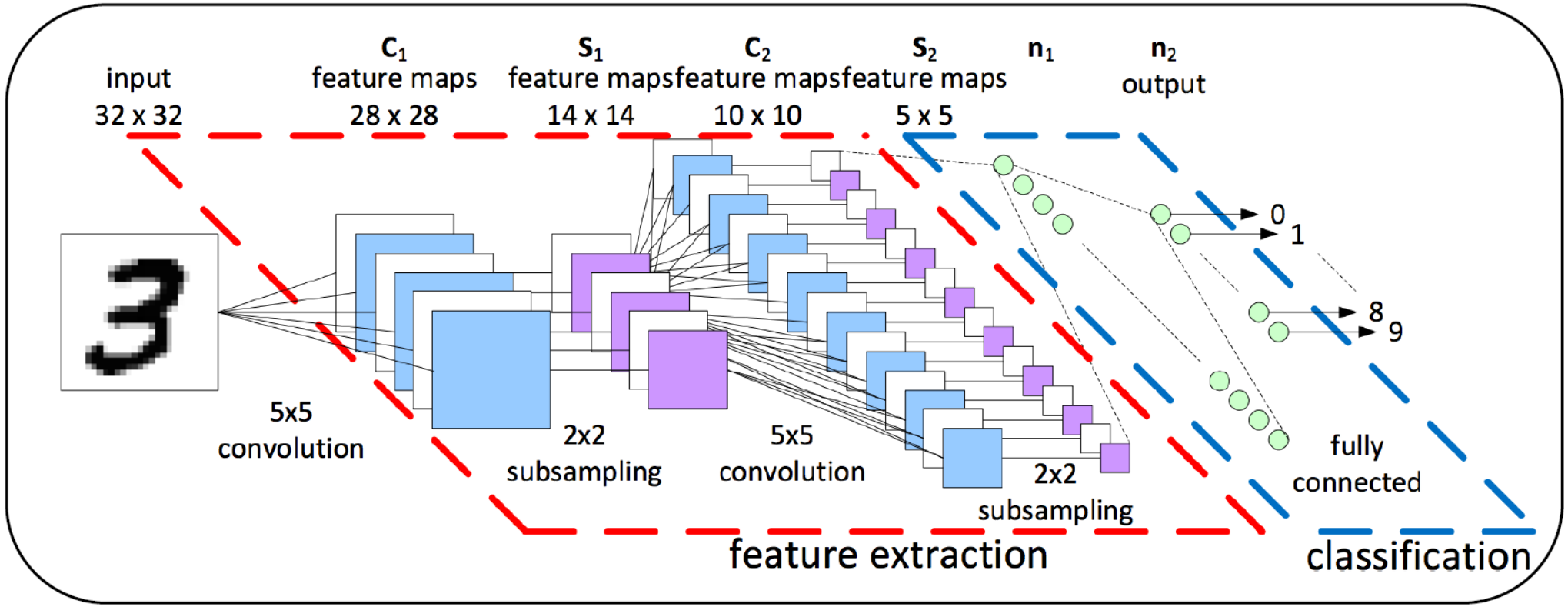
Layers in CNNs



MLP



CNNs


input
32 x 32
C1
feature maps
28 x 28
S1
feature maps
14 x 14
C2
feature maps
10 x 10
S2
feature maps
5 x 5
n1
n2
output
0
1
8
9
5x5
convolution
2x2
subsampling
5x5
convolution
2x2
subsampling
fully
connected
feature extraction
classification